

Independent Project

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Abstract

There is currently no perfect model for normal changes in brain structure during healthy aging. Rodents are a popular test subject in neurodegenerative disease studies. However, not much is known about the processes of normal aging in rodents. This study aims to evaluate the volume of four brain regions: corpus callosum, anterior commissure, amygdala, and dentate gyrus across young and old populations of mice using stereology of Nissl-stained brain slices. 10 Wild-type mice were selected for this study. 5 of the mice were 148 days old and constitute the young population while the other 5 were 375 days old. We found no significant differences in the regions of interest outlined above between the young and old subjects. These results are consistent with other papers describing the aged differences in hippocampal volume among mice. This indicates a need for better healthy-aging mice models to be used as controls in neurodegeneration experiments. A lack of understanding regarding normal mouse aging may lead to faulty claims being made regarding neurodegenerative disease experiments.

Introduction

Human life-expectancy has dramatically increased with medical advances over the past few centuries. This coupled with changes in economic systems which encourage families to have fewer children has led in a shift to a older population. The United States Census Bureau states that there will be a 6% increase in the total number of adults 65 and older between 2018 and 2030 (Vespa, 2018). As the risk to develop neurodegenerative diseases such as Alzheimer's and Dementia increase with age (Ridderinkhof et al., 2022) the need for neurodegenerative research to support our older populations also increases.

Normal aging still involves degeneration of cognitive function. Throughout life neurons are subjected to degradation through mitochondria dysfunction, excitotoxicity, DNA damage, cellular waste, oxidative stress, inflammation and many other molecular mechanisms (Lee et al., 2022). However, normal aging should not affect severely impact an individual's quality of life the way a neurodegenerative disease such as Alzheimer's does. Current studies have shown little recourse once Alzheimer's has been identified. However, treatments have been used to slow down neuronal death caused by disease (Lo et al., 2017).

The amygdala is implicated in reversal learning tasks and fear. A study showed that social isolation in older mice exacerbated necroptosis of the amygdala leading to impairments in learning (Zhang et al., 2022). Older populations are more likely to become isolated as deteriorating health prevents them from participating in social activities. The projection between the dentate gyrus and the hippocampus are important in memory. Volume loss of the dentate gyrus in humans has been associated with impairment in performing memory retrieval tasks (Hayek et al., 2020). The corpus callosum is necessary to perform intra-hemispherical tasks such as responding to a visual stimulus on the opposite side of the body. A study analyzing the response to interhemispheric tasks between old and young populations found its function to be maintained throughout old age even if reaction time is slowed (Reuter et al., 2000).

Stereology is a method to calculate the volume of a structure using Cavalieri's principle to overlay 2D slices and calculate the distance in-between. This methodology has had increasing popularity for analyzing the aging brain because it reduces biases introduced by cell counting (Schmitz et al., 2007) and allows for quicker analysis of the regions of interest.

One of the most common animal models used in research is the mouse. In order to conduct proper experiments a control of normal aging mice must be developed. Currently, there is a lack of research regarding the normal processes of aging within mice. This study analyzes the volume of the corpus callosum, anterior commissure, dentate gyrus, and amygdala in wild type mice across an old and young population. If wild-type mice are a sufficient model for normal aging then, changes in brain structure should model the same changes that occur with healthy human aging.

Methods

Animals

10 Wild-type mice were selected for this experiment. 5 of the mice were 375 days old 2 of which were male and 3 were female. The remaining 5 mice were 148 days old 2 males and 3 females. Using 2.5% Avertin IP (CAS No. 75-80-9; Sigma Aldrich) mice were anaesthetized before performing trans cardiac perfusion utilizing phosphate buffered saline (MDL MFCD00131855; Sigma Aldrich) and 4% paraformaldehyde (MDL MFCD00133991; Sigma Aldrich). Afterwards, their brains were removed and placed in a solution containing 4% PFA and PBS to rest for 24 hours. For long-term storage, the brains were moved to a 30% sucrose, 0.01% sodium azide (CAS No. 26628-22-8; Sigma Aldrich) cryoprotectant solution and then frozen.

Nissl Stain

Mice brains were removed from cryoprotectant and sliced at 30 μm with an 860 American Optical Company sliding microtome. First the brass stage of the microtome was leveled using a precision bubble leveler. Then dry ice was added to the wells of the stage and sprayed with 70% EtOH to freeze the stage. A transfer pipette containing 0.1 M PB was

dispersed to create a small platform on the stage where the brain would later be mounted. Blades of the microtome were cleaned using 100% EtOH and deionized water. The stage of 0.1 M PB was leveled by sliding the microtome blade across the surface until flat. A razor blade was used to remove the posterior portion of the cerebellum and medulla of the mice's brain to provide a flat surface for mounting. Then the brains were set on the microtome's stage with the corpus callosum facing the user and olfactory bulbs facing upwards. The brains were coronally sliced in full and held in 12-well plates containing cryoprotectant for protection of the tissue. Afterwards, sections spanning bregma 1.96 to -4.03 containing the regions of interest: corpus callosum, anterior commissure, amygdala, and dentate gyrus were flow-mounted onto charged slides and dried for 24 hours to prepare for Nissl staining.

First the tissue was gradually dehydrated moving through 2-minute washes of ethyl alcohol diluted with deionized water at concentrations of: 50%, 70%, 95%, 100% and an additional wash of 100% EtOH. Then, tissue was de-fatted using Citrisolv (Catalog# 91547; Fisher Scientific, Fair Lawn, NJ). Afterwards, the slides were re-hydrated in baths of EtOH for 2-minute washes in the reverse of the dehydration step at concentrations of: 100%, 100%, 95%, 70%, and 50%. Finally, tissue was stained for 10 seconds in a thionine solution comprised of 113mL of 0.2M acetic acid (Catalog# 64-19-7; Fisher Scientific, Fair Lawn, NJ), 85mL of sodium acetate (Catalog# 127-09-3; Fisher Scientific, Fair Lawn, NJ), and 0.25 grams of 80% thionine powder (Catalog# 581-64-6; Fisher Scientific, Fair Lawn, NJ) until reaching a pH of 4.5. A few drops of deionized water were added as needed for rehydration of tissue. Slides were left in the biosafety cabinet with the fan on to dry before visualization.

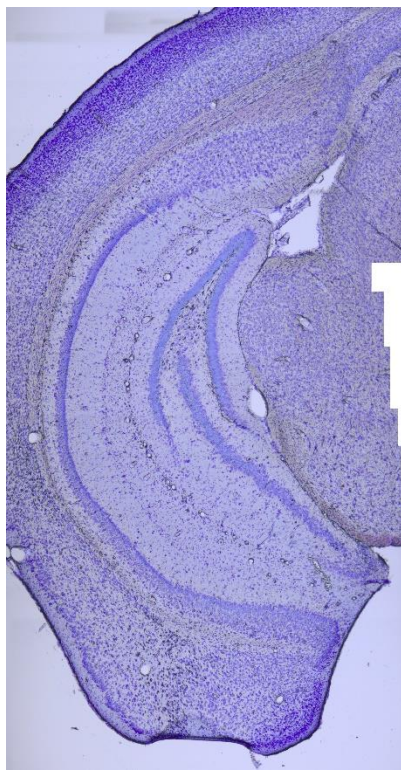
Data analysis

Slides were visualized using a Zeiss axioscope on GRYPHAX software. The panoramic feature was used to capture the entire slice of brain tissue. Images were then compiled into an online database and analyzed using Image J. The regions of interest were measured in pixels and then converted to millimeters. Distance between slices was accounted for when calculating the total volume of each brain region. Using the following equation: $\text{Pixels} \times 0.00074 \times \text{distance from last slice}$. Then a student's t test comparing the volumes of each region of interest between old and young was run through JASP.

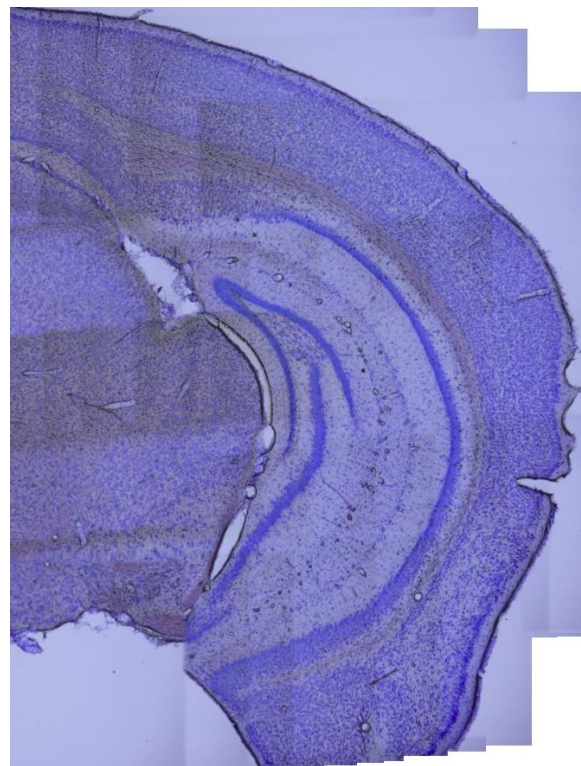
Results

Criteria for significance was set at $P \leq 0.05$. The mean volume of the anterior commissure was not shown to be significant between old and young populations ($P=0.143$, $t=-1.625$). The young group was shown to have a higher volume greater than one standard deviation (Cohen's $d= -1.028$) with the young population averaging at a mean of 632.834 mm^3 ($SE= 52.210$) and the old population averaging at a mean of 452.961 mm^3 ($SE= 97.611$). The mean volume of the corpus callosum was not shown to be significant between old and young populations ($P=0.258$, $t=-1.217$). The young group was shown to have a higher volume greater than one half of a standard deviation (Cohen's $d= -0.769$) with the young population averaging at a mean of 5798.531 mm^3 ($SE= 1465.504$) and the old population averaging at a mean of 3981.323 mm^3 ($SE= 289.107$). The mean volume of the amygdala was not shown to be significant between old and young populations ($P=0.74$, $t=-2.054$). The young group was shown to have a higher volume greater than one standard deviation (Cohen's $d= -1.299$) with the young population averaging at

a mean of 1493.204 mm^3 (SE= 450.513) and the old population averaging at a mean of 534.102 mm^3 (SE= 122.860). The mean volume of the dentate gyrus was not shown to be significant between old and young populations ($P=0.807$, $t=-0.252$). The young group was shown to have a higher volume less than one standard deviation (Cohen's $d= -0.160$) with the young population averaging at a mean of 3283.141 mm^3 (SE= 848.750) and the old population averaging at a mean of 3047.722 mm^3 (SE= 388.485).



A.



B.

Figure 1. Representative Nissl stain at Bregma -3.27 mm. (A) Is a slice from a young mouse containing corpus callosum and dentate gyrus. (B) Is a slice from an old mouse containing the same regions of interest.

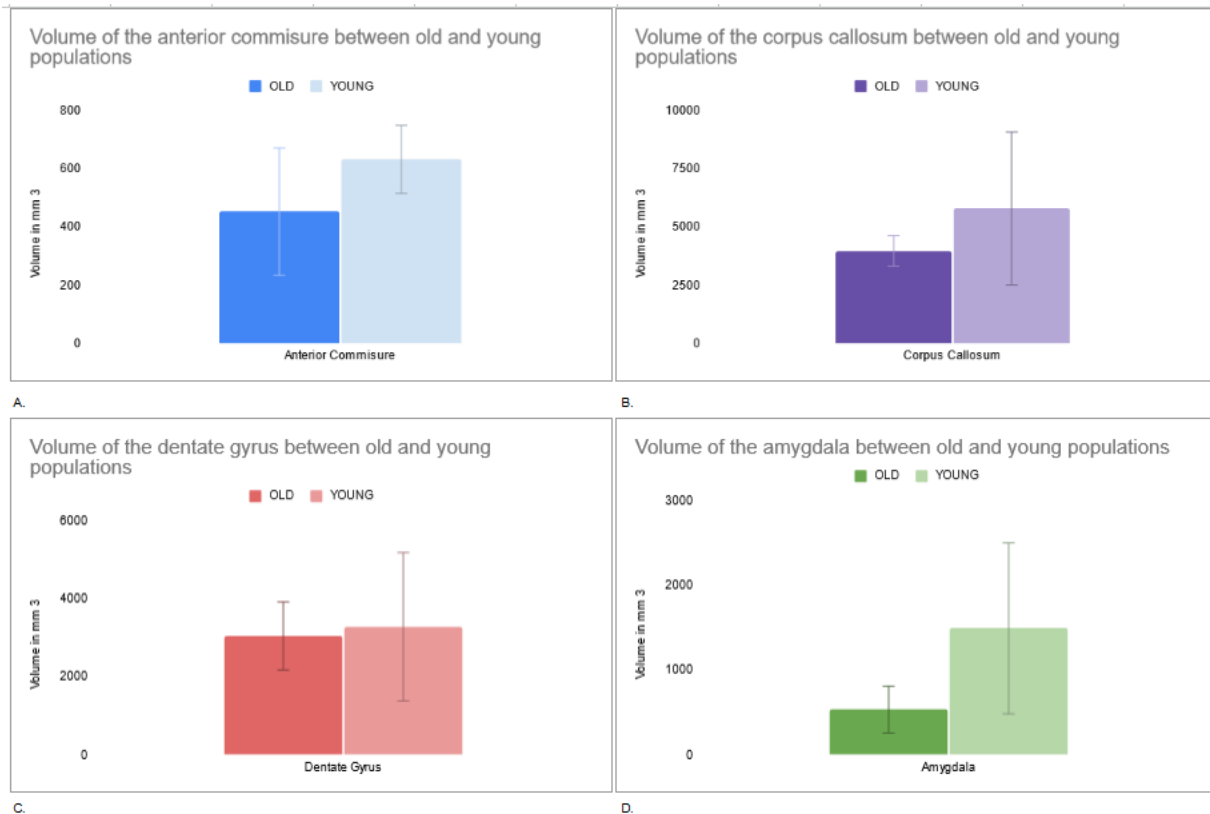


Figure 2. Graphs of the volume of a region of interest between old and young populations.

(A) Volume of the anterior commissure between old (SD= 218.266) and young (SD= 116.746) show a large overlap. (B) Volume of the corpus callosum between old (SD= 646.463) and young (SD= 3276.966) show a large overlap. (C) Volume of the dentate gyrus between old (SD= 868.678) and young (SD= 1897.863) show a large overlap. (D) Volume of the amygdala between old (SD= 247.724) and young (SD= 1007.377) show a large overlap.

Discussion

This study utilized stereology on Nissl-stained mice brain tissue in order to compare the volumes of the corpus callosum, anterior commissure, amygdala, and dentate gyrus across old and young populations. The results of this staining showed no significant difference of the regions of interest between the populations. One explanation for this is that sub-regions of the

area of interest are affected differently throughout the brain. A study analyzing the corpus callosum in adults showed that older adults show increased axon diameter within the genu of the corpus callosum but does not continue to the splenium (Fan et al., 2019). There are also reported sex differences in this area, females have shown larger myelinated callosal fibers in the midbody of the corpus callosum (Aboitiz et al., 1996). The same implications of aging on the corpus callosum are supported for the amygdala. A study using MRI to image the amygdala found decreases in volume with age in most of the amygdala sub-nuclei but only in males (Arash et al., 2019). They also discovered differences within sub-regions of the amygdala, the central nucleus of the amygdala was not affected by aging (Aghamohammadi et al., 2019). Another study found that older males have a significantly longer anterior commissure but not females (Lee et al., 2008). These results are in direct opposition to the results of this study the greater number of female participants may account for this. Previous research indicates a significant decrease in dentate gyrus volume with aging (Feng et al., 2020).

However, evidence that regions affected by normal human aging are not affected in mice. In a study of 52 mice with spatial memory deficits no significant changes to hippocampal volume were detected (Rasmussen et al., 1996). There is a lack of research examining the volumetric changes of the regions of interest in this study. These results further support the disparity between human and mice aging.

Another explanation for the variance in data is that the blade angle on the microtomes was not consistent throughout slicing. Some mice brains appear to have one hemisphere in a completely different bregma based on the changes in landmark features. To further exacerbate this problem, the orientation of slices was not regulated meaning that any slice could have flipped over during mounting showing the right hemisphere on the left or vice versa. During

mounting it is also possible to stretch and tear the tissue which will result in false volumetric differences. Panoramic photography on the Zeiss Axioscope is performed by rapidly taking images while the stage is moved and merging them together. This may result in inconsistent expansion of the regions of interest. The method used to perform stereology was highly subjective, differences in outlining regions of interest between researchers may also contribute to the large variance of data. To improve on this project, it would be ideal for all researchers to take measurements of the regions of interest for each brain and then take the mean of these values for further calculations however, time constraints prevented more thorough work. Image J is also not the ideal software to perform this type of experiment. A software such as Stereo Investigator could assist in keeping the measurements regulated.

This study demonstrates the need for better healthy aging mice models so that research can be applied to normal human aging. Without the availability of these models, studies regarding aging and neurodegeneration in mice will be less translational to our population.

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